

Value of gastrografen in adhesive small bowel obstruction after unsuccessful conservative treatment: a prospective evaluation.

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AIM: Gastrografen is a hyperosmolar water-soluble contrast medium. Besides its predictive value for the need for operative treatment, a potential therapeutic role of this agent in adhesive small bowel obstruction has been suggested. This study aimed at evaluating the effectiveness of gastrografen in adhesive small bowel obstruction when conservative treatment failed.

METHODS: Patients with adhesive small bowel obstruction were given trial conservative treatment unless there was fear of bowel strangulation. Those responded in the initial 48 h had conservative treatment continued. Patients who showed no improvement in the initial 48 h were given 100 mL of gastrografen through nasogastric tube followed by serial abdominal radiographs. Patients with the contrast appeared in large bowel within 24 h were regarded as having partial obstruction and conservative treatment was continued. Patients in which the contrast failed to reach large bowel within 24 h were considered to have complete obstruction and laparotomy was performed.

RESULTS: Two hundred and twelve patients with 245 episodes of adhesive obstruction were included. Fifteen patients were operated on soon after admission due to fear of strangulation. One hundred and eighty-six episodes of obstruction showed improvement in the initial 48 h and conservative treatment was continued. Two patients had subsequent operations because of persistent obstruction. Forty-four episodes of obstruction showed no improvement within 48 h and gastrografen was administered. Seven patients underwent complete obstruction surgery. Partial obstruction was demonstrated in 37 other cases, obstruction resolved subsequently in all of them except one patient who required laparotomy because of persistent obstruction. The overall operative rate in this study was 10%. There was no complication that could be attributed to the use of gastrografen.

CONCLUSION: The use of gastrografen in adhesive small bowel obstruction after unsuccessful conservative treatment is safe and reduces the need for surgical intervention.